

Market Microstructure

INTRODUCTION

Market microstructure is the study of financial markets and how they operate. Market microstructure research primarily focuses on the structure of exchanges and trading venues (e.g. displayed and dark), the price discovery process, determinants of spreads and quotes, intraday trading behavior, and transaction costs. Market microstructure continues to be one of the fastest growing fields of financial research due to the rapid development of algorithmic and electronic trading.

As a result of the high speed evolution of financial markets in the last decade, today's number of trading venues and exchanges have mushroomed, with more on the horizon, and trading processes have evolved exponentially in complexity and sophistication. Traditional trading functions and participants have been surpassed by computers and electronic trading agents. Human intervention in the trading process has expanded from the traditional matching and routing of orders to one requiring complex analysis and sophisticated real-time decision making. The market microstructure analyst is now tasked with understanding all issues surrounding the ever-changing marketplace.

Most of the academic research up until now has focused on valuation techniques to uncover the fair market price of an instrument, expected return forecasts, and risk modeling techniques. Analysts may employ a bottom-up approach to determine the fair value price of a company by examining the company's balance sheet, fundamentals, sales figures, year-over-year growth, and revenue forecasts. Analysts may also utilize a top-down approach to determine which factors, sectors, or other subgroups will likely under- or overperform going forward. Additionally, analysts may utilize a quantitative process, such as the capital asset pricing model (CAPM) or arbitrage pricing theory (APT) to perform asset allocation and determine optimal portfolio mixes.